

Total No. of Questions : 8]

SEAT No. :

PC2483

[Total No. of Pages : 5

[6354]-612

B.E. (Mechanical)

**HEATING VENTILATION AIR-CONDITIONING AND
REFRIGERATION**

(2019 Pattern) (Semester - VII) (402041)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary
- 3) Figures to the right indicate full marks
- 4) Use of Scientific Calculator is allowed
- 5) Assume Suitable data if necessary

Q1) a) Explain with neat sketch the flooded type evaporator. **[6]**

b) Explain with neat schematic diagram the frost control circuit used in VCR cycle. **[5]**

c) Explain with neat schematic CO₂ trans critical cycle. **[6]**

OR

Q2) a) Explain with a neat sketch Thermostatic Expansion Valve. **[6]**

b) Explain with neat schematic Simple Ejector Refrigeration System. **[5]**

c) Discuss the following terms used in thermodynamics analysis of Simple Ejector Refrigeration Cycle. **[6]**

i) Entrainment Ratio

ii) Entrainment efficiency

iii) Nozzle Efficiency

P.T.O.

- Q3) a)** Explain the terms: [8]
- GRSHF
 - BPF
 - ERSHF and
 - SHF
- b)** A conference room for sitting 100 persons is to be maintained at 22°C DBT and 60% relative humidity. The outdoor conditions are 40°C DBT and 27 °C WBT. The various loads in the auditorium are as follows: [10]
- Sensible and latent heat loads per person 80 W and 50 W respectively;
 - Lights and fans, 15000 W;
 - Sensible heat gain through glass ceiling etc. 15000 W.
 - The air infiltration is 20 m³/min and fresh air supply is 100 m³/min.
 - Two-third of recirculated room air and one-third of fresh air are mixed before entering the cooling coil.
 - The bypass factor of the coil is 0.1.

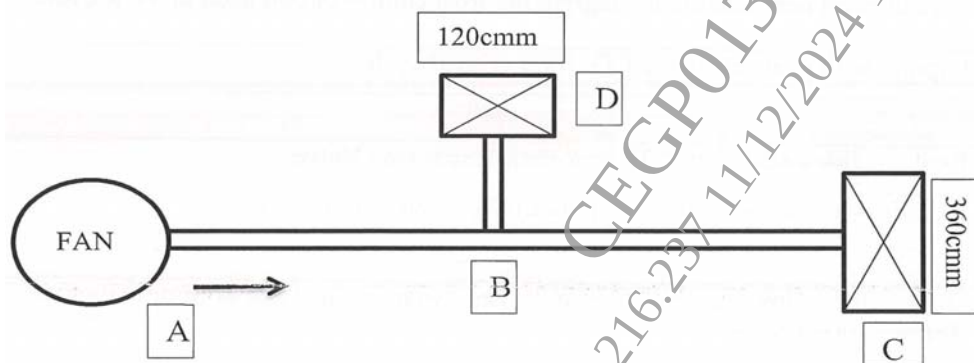
Determine

- Apparatus dew point,
- Grand total heat load and
- Effective room sensible heat factor

OR

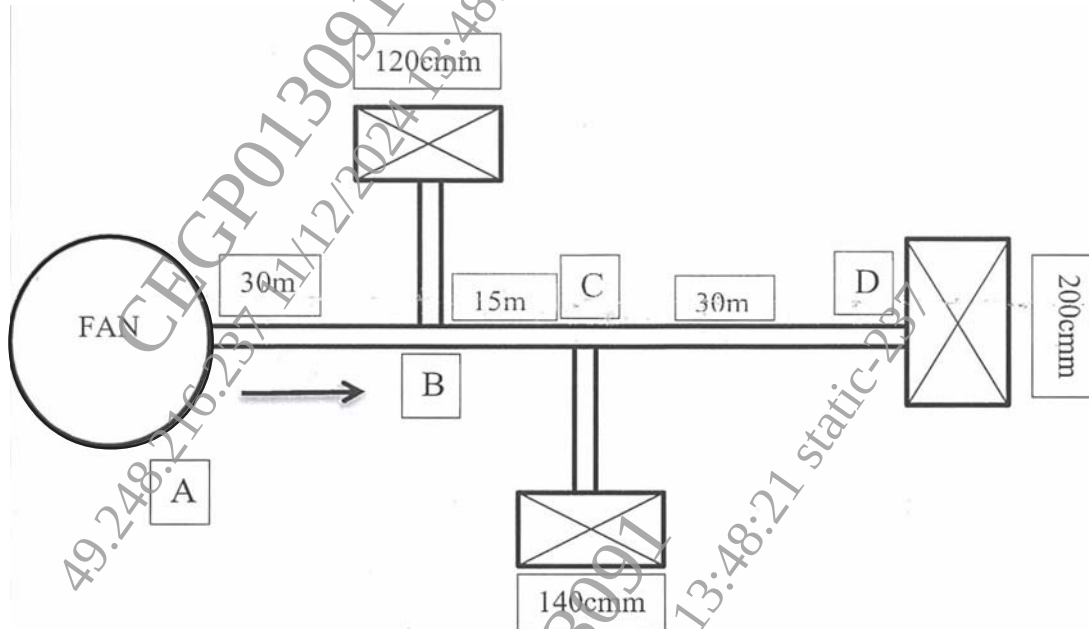
- Q4) a)** Explain in detail the “decrement factor” and “time lag”. [8]
- b)** The air-handling unit of an air-conditioning plant supplies a total of 4500 cmm of dry air which comprises by weight 20% fresh air at 40°C DBT, 27°C WBT, and 80 % recirculated air at 25°C DBT and 50% RH. The air leaves the cooling coil at 13°C saturated state. Calculate the following. [10]
- Total cooling load, and
 - Room heat gain

- Q5) a)** Explain Natural Ventilation and Mechanical Ventilation? [8]
- b)** Using Equal friction method, determine the duct diameter and velocity for section AB, BD and BC. Assume velocity in the main duct AB=600 m/min. Also Calculate maximum pressure drop in the duct system. Distance AB = 30 m, Distance BC = 30 m and Distance BD=10 m. Refer the figure as given below. [10]



OR

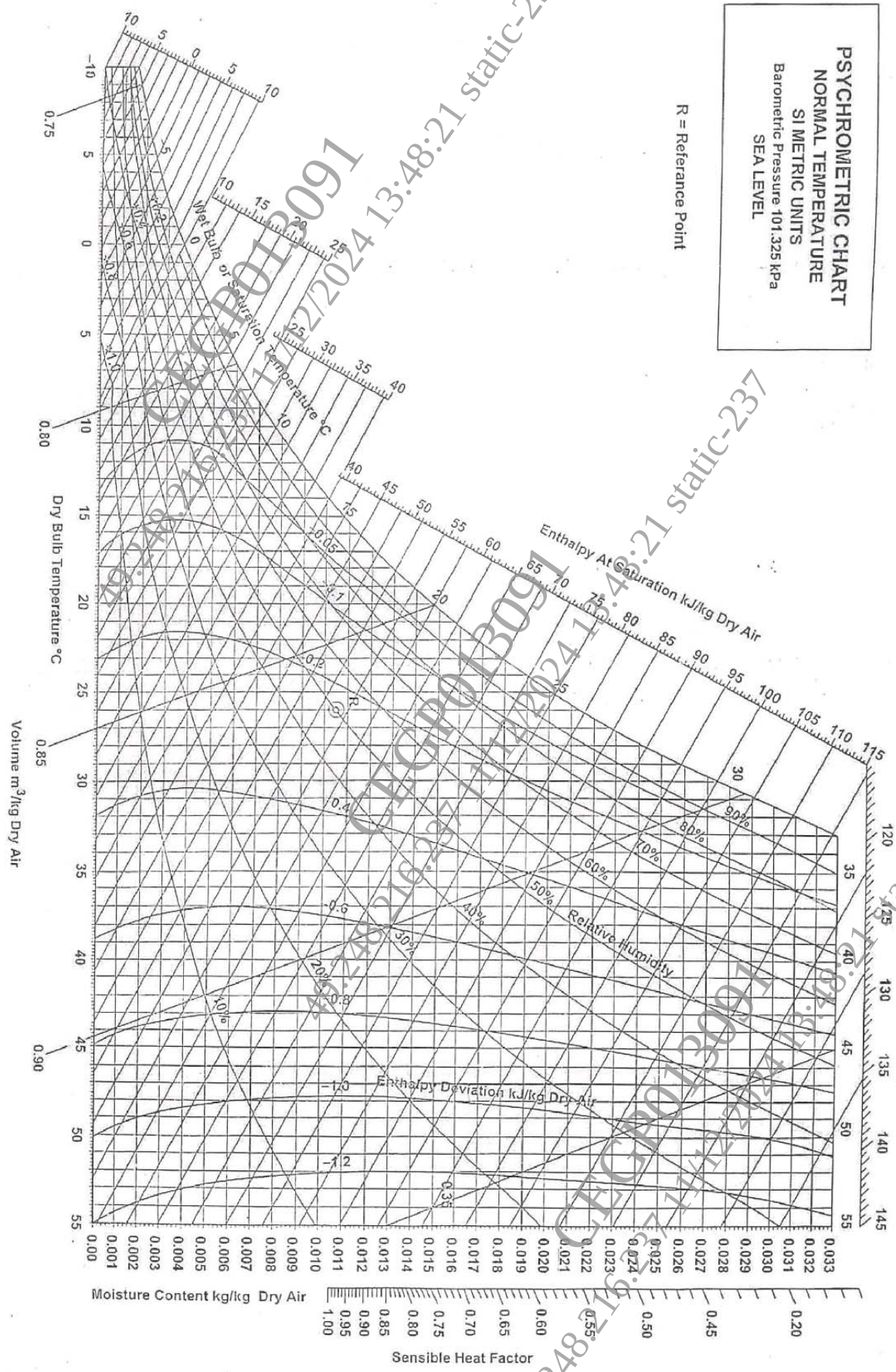
- Q6) a)** Define the following as applied to “Air Distribution System” Intake, Outlet, Grille, Register, Diffuser, Filter, Throw and Primary Air. [8]
- b)** Using equal Friction pressure drop method and friction chart estimate the diameter and velocity pressure in AB, BC, CD, BE, CF Duct sections. Take the assumption if any. [10]



- Q7) a)** Explain with neat sketch winter Air conditioning system. [6]
- b)** Draw and Explain water to water heat pump circuit. [6]
- c)** Write a short note on solid packed tower. [5]

OR

- Q8) a)** Explain with neat sketch variable refrigerant flow (VRF) air conditioning system. [6]
- b)** Explain with neat sketch all year round air conditioning system. [6]
- c)** State the factors to be considered while selecting a system of air conditioning. [5]



Friction Chart for Circular Ducts

